

```
> db.books.aggregate([
  {
    $group:{
      _id:"$title",totalprice:{$sum:"$price_per_book"}
    }
  },
  {$sort:{totalprice:1}}
])
< {
  _id: 'Basic for Beginners',
  totalprice: 40
}
{
  _id: 'Computer Studies',
  totalprice: 75
}
{
  _id: 'Dbase Dummies',
  totalprice: 130
}
{
  _id: 'DOS Guide',
  totalprice: 175
}
{
  _id: 'Guide Network',
  totalprice: 200
}
```

```
> db.books.aggregate([
  {
    $group:{
      _id:"$edition",
      totalprice:{$sum:"$price_per_book"}
    }
  }
])
< {
  _id: 'Second Edition',
  totalprice: 545
}
{
  _id: 'First Edition',
  totalprice: 562
}
{
  _id: 'Third Edition',
  totalprice: 1575
}
```

```
> db.books.aggregate([
  {
    $group: {
      _id: "$title",
      totalAmount: { $sum: { $multiply: ["$quantity_in_stock", "$price_per_book"] } }
    }
  },
  {
    $sort: { totalAmount: 1 }
  }
]);
< {
  _id: 'Basic for Beginners',
  totalAmount: 120
}
{
  _id: 'Computer Studies',
  totalAmount: 150
}
{
  _id: 'Mastering Windows',
  totalAmount: 225
}
{
  _id: 'DOS Guide',
  totalAmount: 525
}
{
  _id: 'Guide Network',
```

```
> db.purchase.aggregate([
  {
    $group: {
      _id: "book_id",
      averagePurchaseAmt: { $avg: "$purchase_amt" }
    }
  },
  {
    $project: { _id: 0, averagePurchaseAmt: 1 }
  }
]);
< {
  averagePurchaseAmt: 679
}
day4_asg>
```

```
> db.books.aggregate([
  {
    $group: {
      _id: "$title",
      totalAmount: { $sum: { $multiply: ["$quantity_in_stock", "$price_per_book"] } }
    }
  },
  {$match:{totalAmount:{$gt:1000}}},
  {
    $sort: { totalAmount: 1 }
  }
]);
< {
  _id: 'Turbo C++',
  totalAmount: 1350
}
{
  _id: 'Advanced Pascal',
  totalAmount: 1400
}
{
  _id: 'COBOL',
  totalAmount: 4000
}
```

```
> db.purchase.aggregate([
  {
    $lookup: {
      from: "salesman",
      localField: "salesman_id",
      foreignField: "_id",
      as: "salesman_details"
    }
  },
  {
    $lookup: {
      from: "customer",
      localField: "customer_id",
      foreignField: "_id",
      as: "customer_details"
    }
  },
  {$unwind:"$salesman_details"},
  {$unwind:"$customer_details"},
  {$project:{
    _id:0,
    customerName:"$customer_details.first_name",
    salesman:"$salesman_details.name"
  }}
])
< {
  customerName: 'Gordon',
  salesman: 'Nail Knite'
```

```
> db.customer.aggregate([
  {$match:{state:{$eq:"WA"}}},
  {$project:{_id:0,first_name:1,last_name:1,city:1,state:1}},
  {$sort:{city:1}}
])

< {
  first_name: 'Gordon',
  last_name: 'Hee',
  city: 'Bellevue',
  state: 'WA'
}
{
  first_name: 'Kim',
  last_name: 'Ralls',
  city: 'Bothell',
  state: 'WA'
}
{
  first_name: 'Gabe',
  last_name: 'Mares',
  city: 'Edmonds',
  state: 'WA'
}
{
  first_name: 'Sariya',
  last_name: 'Hampa',
  city: 'Everett',
  state: 'WA'
}
```

```

> db.purchase.aggregate([
  {
    $lookup: {
      from: "salesman",
      localField: "salesman_id",
      foreignField: "_id",
      as: "salesman_details"
    }
  }
])
< {
  _id: ObjectId('6a1541602115320e7bbb132d'),
  order_no: 70001,
  book_id: 3,
  item_quantity: 3,
  purchase_amt: 810,
  date: 2022-11-05T00:00:00.000Z,
  customer_id: 3005,
  salesman_id: 5002,
  salesman_details: [
    {
      _id: 5002,
      name: 'Nail Knite',
      city: 'Paris',
      commission: 0.13
    }
  ]
}

```

```
> db.customer.aggregate([
  {
    $lookup: {
      from: "purchase",
      localField: "_id",
      foreignField: "customer_id",
      as: "Purchase"
    }
  },
  {
    $unwind: {
      path: "$Purchase",
      preserveNullAndEmptyArrays: true
    }
  },
  {
    $project: {
      _id: 0,
      FirstName: "$first_name",
      LastName: "$last_name",
      Phone: "$phone",
      Email: "$email",
      item_quantity: "$Purchase.item_quantity",
      purchase_amt: "$Purchase.purchase_amt"
    }
  },
  { $sort: { FirstName: 1 } }
]);
```